Felipe Meneguitti Dias

AI engineering for healthcare

I am a Senior R&D Specialist at Samsung with a PhD from the University of São Paulo. My work spans healthcare AI, signal processing and model validation, from development to deployment.

Selected work

Heart Institute (InCor), Hospital das Clínicas, University of São Paulo Medical School

Gorgona

I led Gorgona, which analyzes ECGs from urgent care units in Salvador on the InCor server and sends Telegram alerts alongside telemedicine.

ECG classification at InCor

At InCor, I developed models to classify emergency ECGs. I coordinated ECG annotation with cardiologists and worked on integration with the hospital system.

RPMS

I worked with the team on remote monitoring with watches, from collection protocols to PPG processing and algorithms running on the phone.

Samsung · 2026–present | InCor · 2020–2026

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Gorgona

ECG classification integrated into urgent care

Objective

Gorgona analyzes ECGs recorded at urgent care units and notifies the clinical coordination team when an ECG suggests current of injury. The alert arrives through Telegram while the examination also goes to telemedicine. Clinicians assess the patient and decide on care.

My work

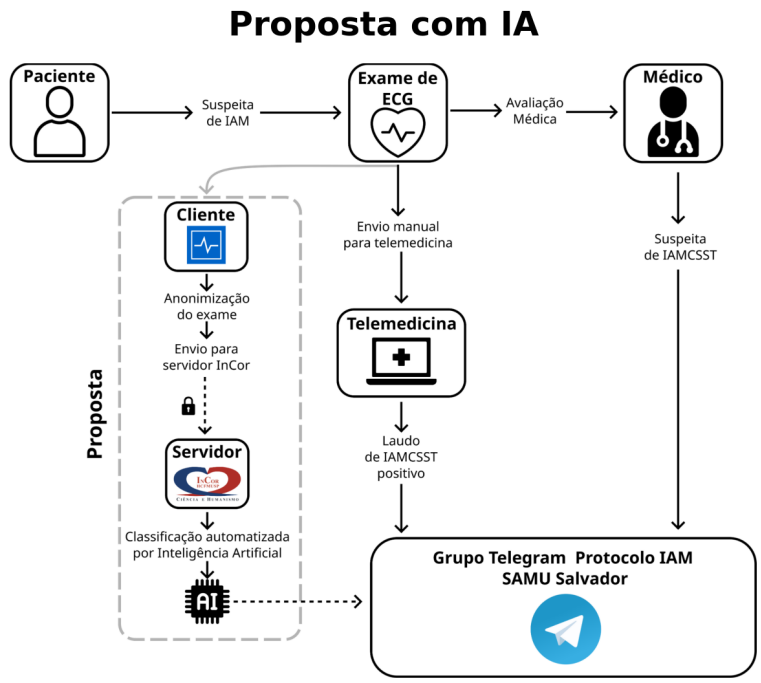
I led the project and developed the initial integration of the model with the Windows application, InCor server, dashboard and Telegram. I worked with the clinical team, who defined the care protocol and reviewed the ECGs.

Previous workflow

Manual submission to telemedicine and staff communication with the coordination team, based on clinical assessment.

With Gorgona

Windows client, configurable anonymization, InCor server and automated Telegram alert.



Proposed AI workflow. Original project figure, in Portuguese.

Telemedicine and clinical decisions remain part of the workflow. HTTPS and per-installation credentials complement configurable local anonymization.

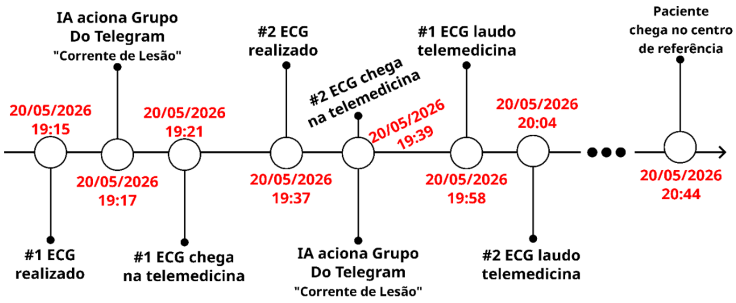
Case study

2 min
ECG to alert in the case

41 min
before the first report

UPA Rodrigo Argolo · 20/05/2026

In the encounter documented on May 20, 2026, at UPA Rodrigo Argolo, admission was at 18:52. Two ECGs were recorded, and each triggered an alert 2 minutes later. Arrival at the referral center was recorded at 20:44.



Case study timeline, starting with the first ECG. Patient identifiers have been removed. Admission was at 18:52.

The first alert arrived 4 minutes before the examination reached telemedicine. The alerts preceded the reports by 41 and 25 minutes. These times show that information reached the coordination team earlier. This single case does not establish that AI improved treatment times or outcomes.

Classification model

We started from the AIMED model. I led the team in the 2024 PhysioNet Challenge, where we placed first in classification. The CinC 2024 paper and my thesis describe the method.

References

Image-Based Electrocardiogram Classification Using Pre-trained ConvNext · Computing in Cardiology, 2024.

Classificação automatizada de imagens de eletrocardiogramas por aprendizado profundo · Universidade de São Paulo · Tese de doutorado, 2025.

Image-based deep learning for emergency electrocardiogram classification · medRxiv · Preprint, 2026.

See the full case study for the location map, Windows client and online platform

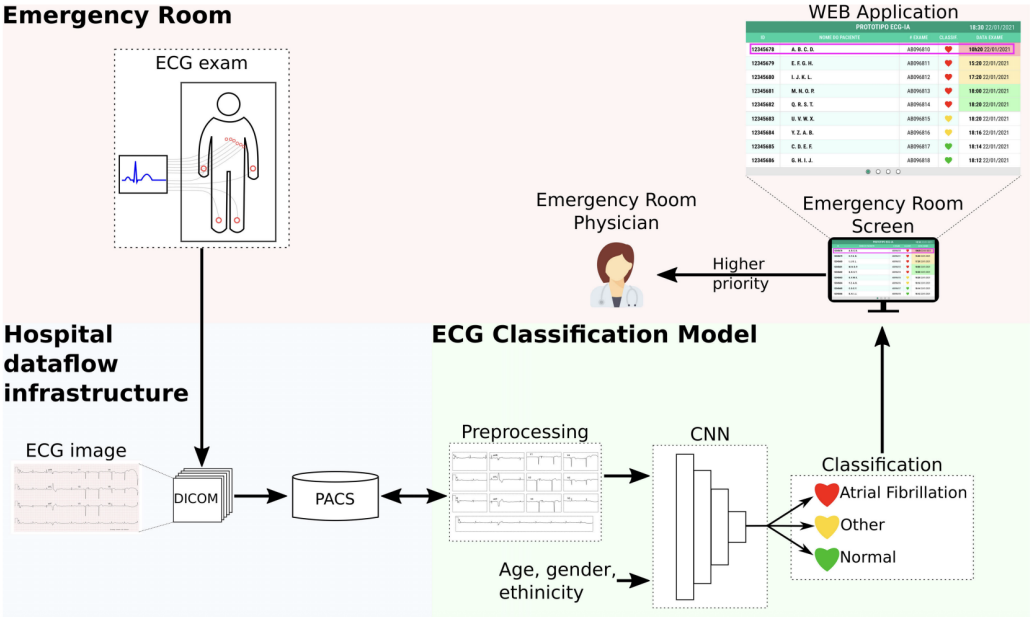
ECG classification at InCor

Objective

The aim was to help the emergency team prioritize ECG review. The classifier presents its analysis in the application clinicians already use at the hospital.

My work

I developed the models and worked on deployment in the emergency department. I also coordinated annotation of around 20,000 ECGs with 19 cardiologists, organizing class definitions, review of disagreements and agreement analysis. After deployment, I worked on cardiologist review and comparison with specialists.



Integration of the ECG, PACS, classifier and emergency department application. Figure 1 from Dias et al., IEEE Access (2023).

Results and evaluation

The 2023 version classified examinations as atrial fibrillation, other or normal. We later worked with a ConvNeXt ensemble for 12 categories. In the 2026 study of 18,519 ECGs from 14,402 patients, held-out macro F1 was 0.807 (95% CI: 0.788–0.825).

Deployment was reviewed by a cardiologist and compared with specialists. The figures above describe the research dataset and test results, rather than the number of examinations processed in routine care.

References

Artificial Intelligence-Driven Screening System for Rapid Image-Based Classification of 12-Lead ECG Exams: A Promising Solution for Emergency Room Prioritization · IEEE Access, 2023.

Image-based deep learning for emergency electrocardiogram classification · medRxiv · Preprint, 2026.

Classificação automatizada de imagens de eletrocardiogramas por aprendizado profundo · Universidade de São Paulo · Tese de doutorado, 2025.

RPMS

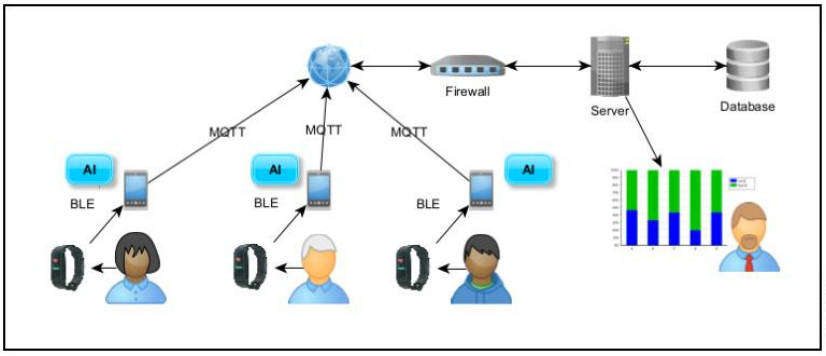
Remote Patient Monitoring System

Objective

RPMS uses watches and a mobile app to track physiological measurements outside the hospital. The phone processes the signals and sends results to a dashboard, where the clinical team can follow each patient.

My work

I worked with prototype watches and helped define how to collect the signals. I organized the data and worked with the team on photoplethysmography (PPG) preprocessing and the project's algorithms.



Watch, app and clinical dashboard. Figure 1 from Moreno, Dias et al., SIoT (2022).

The watch sends PPG to the Android app over Bluetooth Low Energy. The phone processes the signal, estimates blood pressure and heart rate, and sends results over MQTT to a ThingsBoard server. The dashboard displays each patient's measurements over time.

52 volunteers and 15 wristbands in the version published in 2022.

Validation and research

The published protocol keeps participants separate across validation folds. The 2022 implementation estimates blood pressure and heart rate. Signal quality, diabetes and sleep are research areas documented in the papers below.

References

IoT Medical Device Architecture to Estimate Non-invasive Arterial Blood Pressure · Symposium on Internet of Things (SIoT), 2022.

Quality Assessment of Photoplethysmography Signals For Cardiovascular Biomarkers Monitoring Using Wearable Devices · arXiv · Preprint, 2023.

Exploring the limitations of blood pressure estimation using the photoplethysmography signal · Physiological Measurement, 2025.

Machine Learning-Based Diabetes Detection Using Photoplethysmography Signal Features · SBCAS, 2024.

Enhancing Photoplethysmography-Based Sleep Staging Models Through Temporal Context Optimization · Studies in Health Technology and Informatics, 2025.